# CS-663
# Digital Image Processing (DIP)

## Image Enhancement:
## "Point" Operations and Beyond

Suyash P. Awate

# Image Enhancement

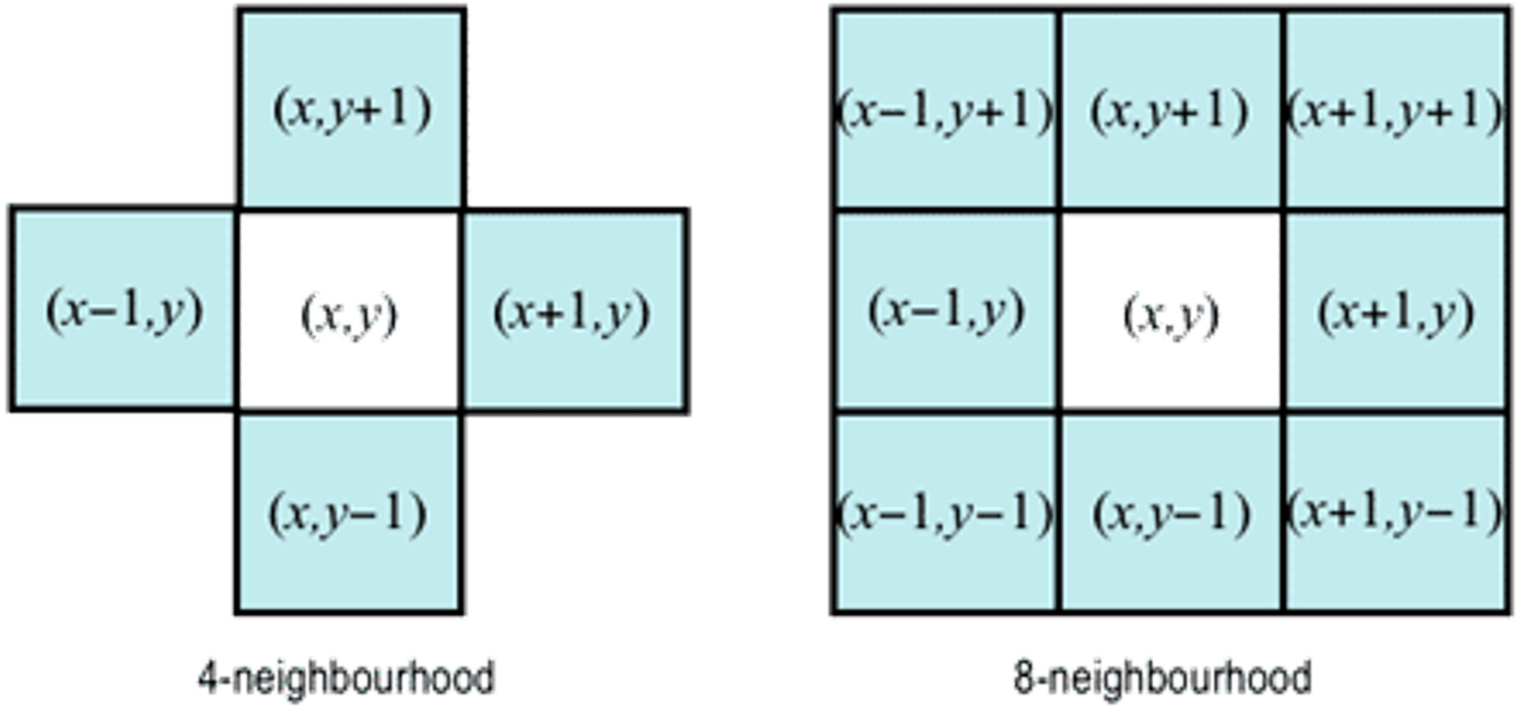
- Image Enhancement: broadly speaking, this involves processing an input image so that output image is more suitable for a specific downstream task
- "Processing" examples
  - Improve contrast, improve sharpness, reduce artifacts like noise and blur, etc.
- "Downstream task" examples
  1) Further post-processing of image, e.g., object detection, image classification
  2) Decision making by experts based on visual interpretation
- Evaluation is two-pronged
  1) Qualitative: visual, subjective
  2) Quantitative: need a ground-truth / gold standard

# "Point" Operations

- Design functions to transform gray levels $f(x)$, at pixel locations $x$, where functions have a single argument which is $f(x)$
- Let input-image intensity at location $x$ be $f(x)$
- Then output-image intensity at location $x$ can be $g(x) = T(f(x))$
- Transformation $T(.)$ can be specified manually, or derived from:
  1) Individual intensities $f(x)$
  2) Intensities $f(y)$ at pixels $y$ in a neighborhood around $x$
  3) All intensities in image

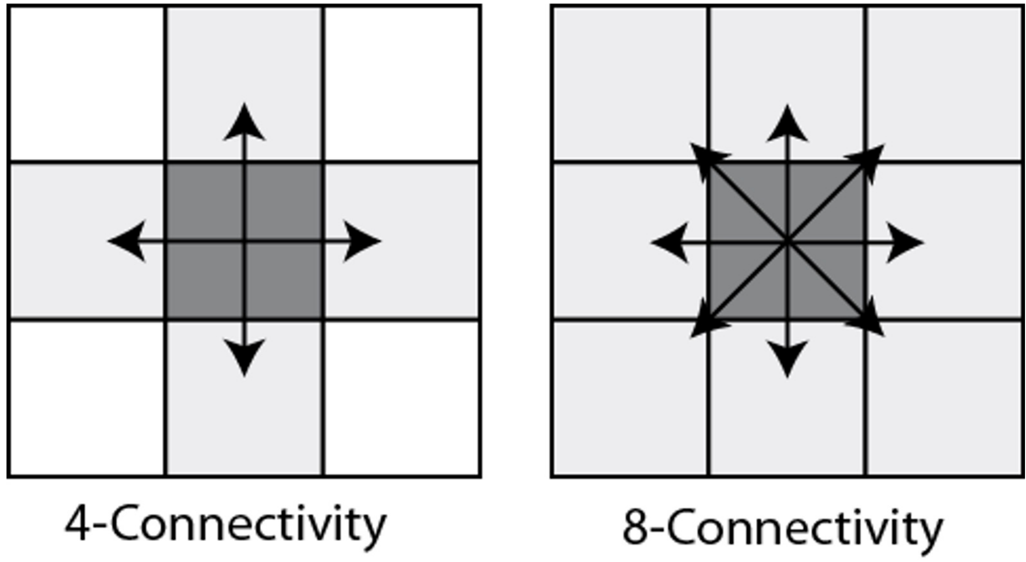
# Neighborhood

- Neighborhood in 2D image
  - 4 neighborhood: 4 closest pixels
  - 8 neighborhood: 8 closest pixels



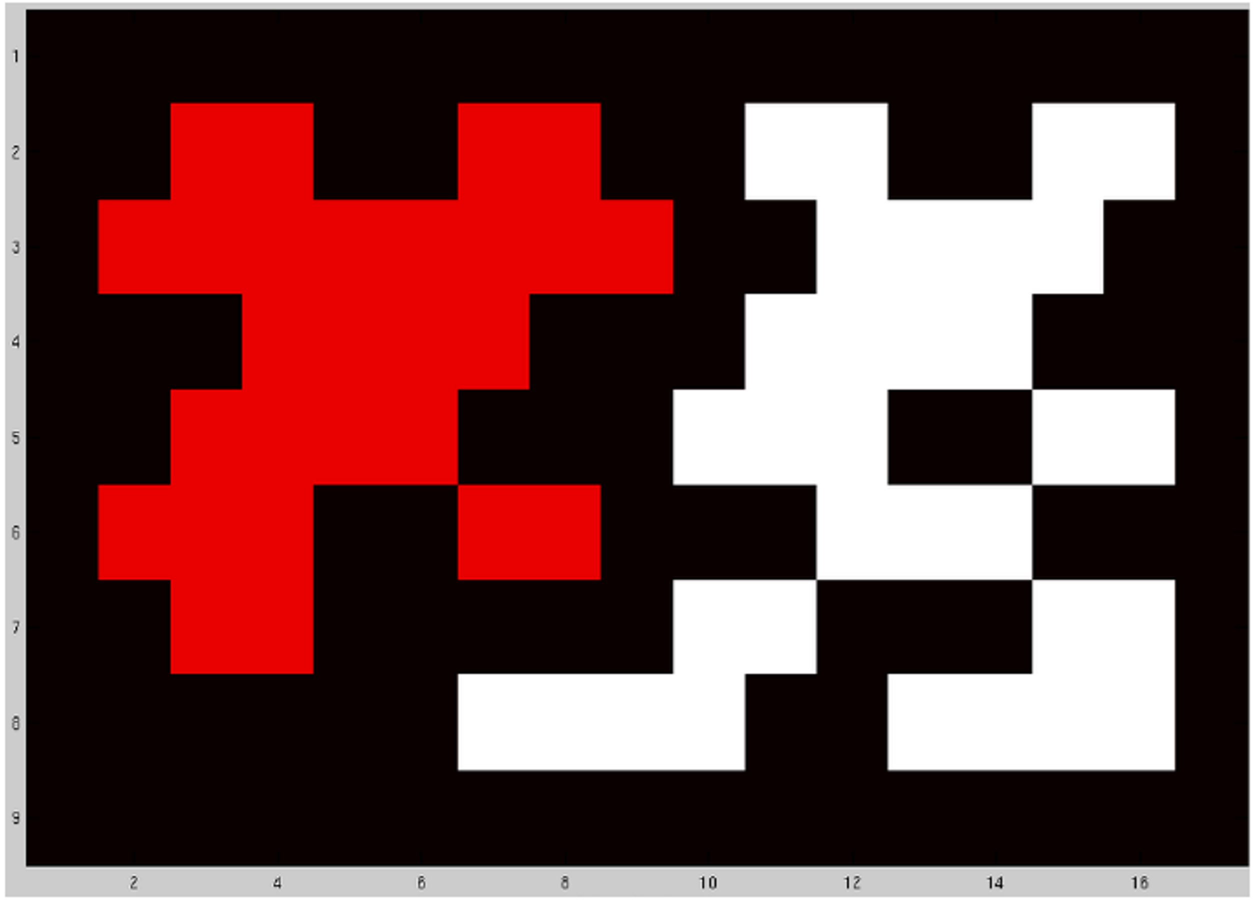

- Related to neighborhood is notion of connectivity
- Connected path between pixels (x,y) and (s,t) = sequence of pixels $(a_1,b_1), (a_2,b_2), \ldots, (a_n,b_n)$ such that:

1) $(a_1,b_1) = (x,y)$
2) $(a_n,b_n) = (s,t)$
3) $(a_{i-1},b_{i-1})$ and $(a_i,b_i)$ are neighbors, for $i=2,\ldots,n$



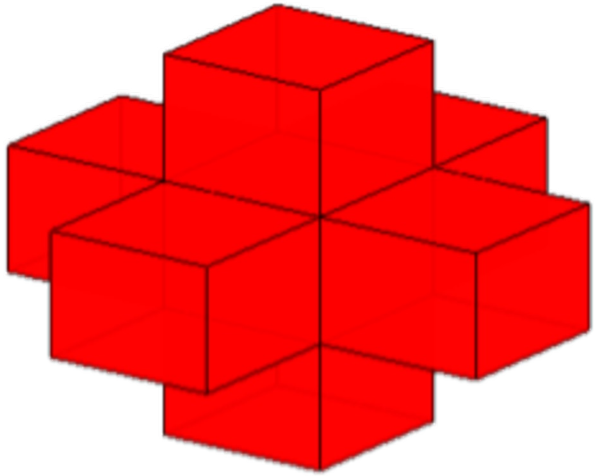
# Neighborhood

- Neighborhood in 2D image
- Connected subset / Region R
  = set of pixels such that
  there exists a connected path between any 2 pixels within R,
  such that path consists entirely of pixels within R
- Is the set of red pixels
  a connected subset
  assuming
  4-neighbor connectivity ?

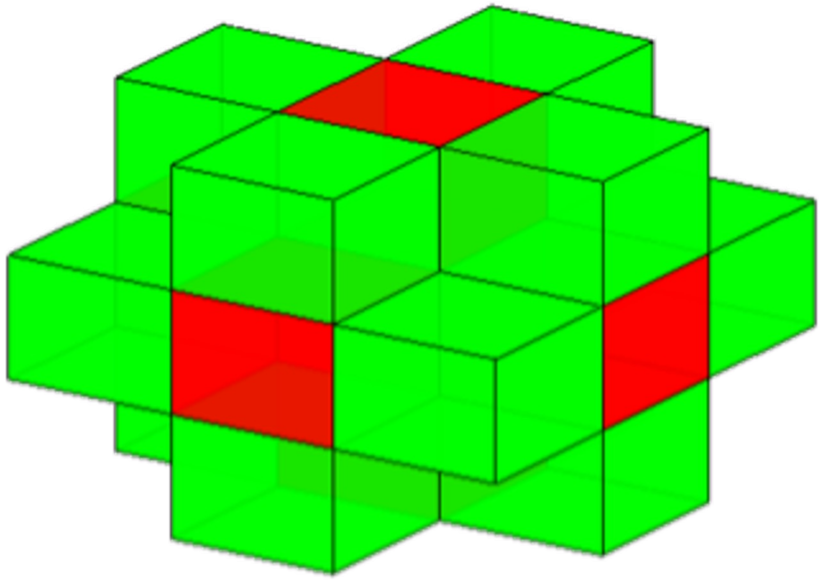


# Neighborhood

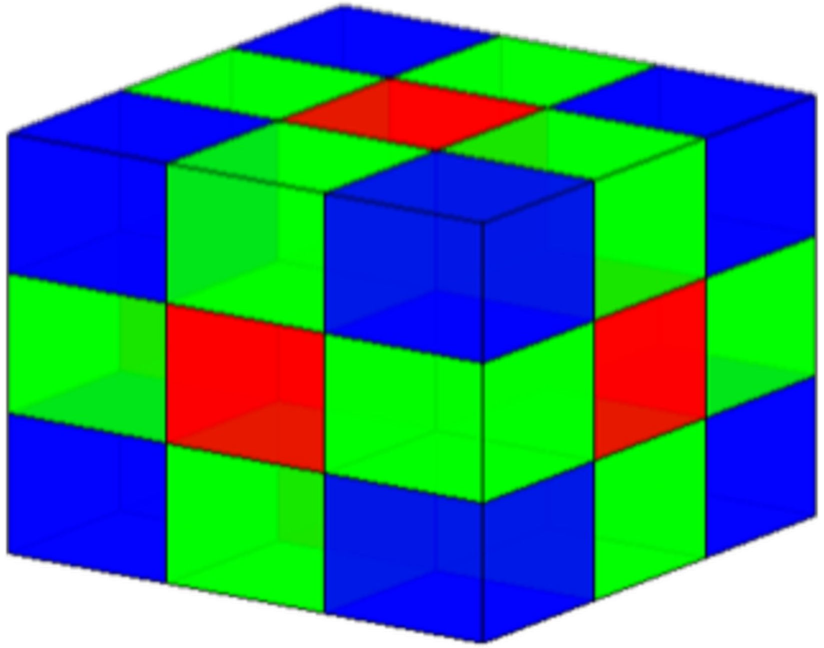
- Neighborhoods in a **3D** image



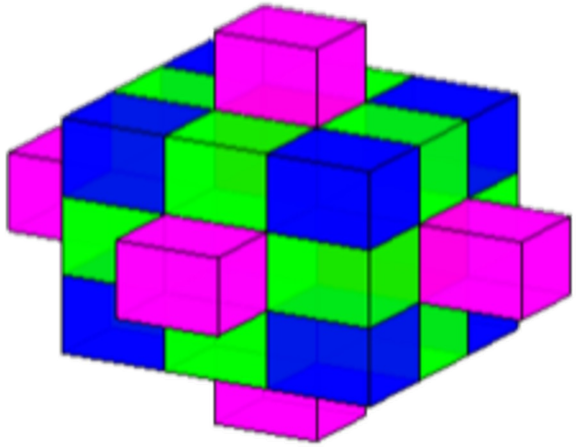
6-Connectivity



18-Connectivity



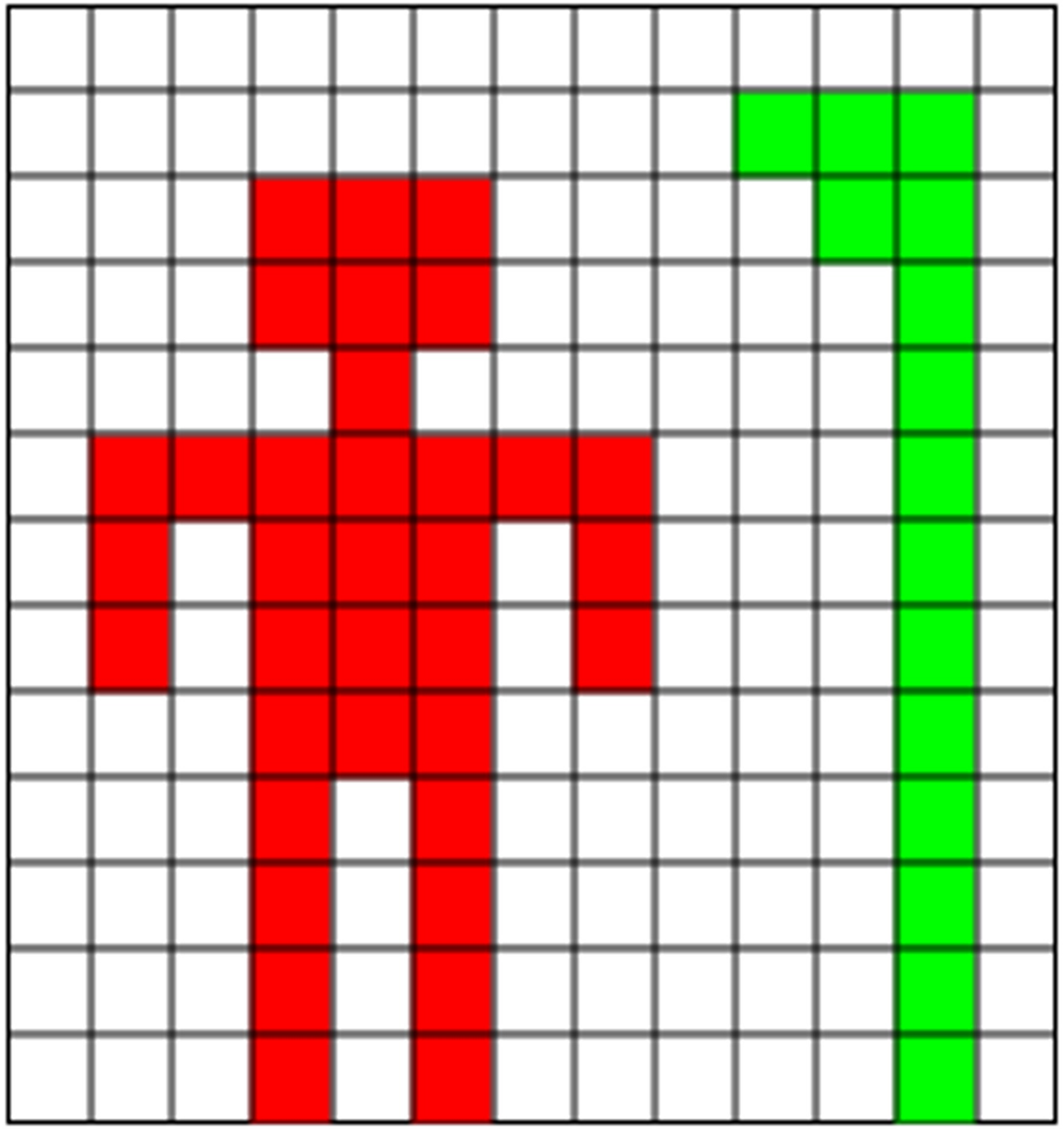
26-Connectivity



32-Connectivity

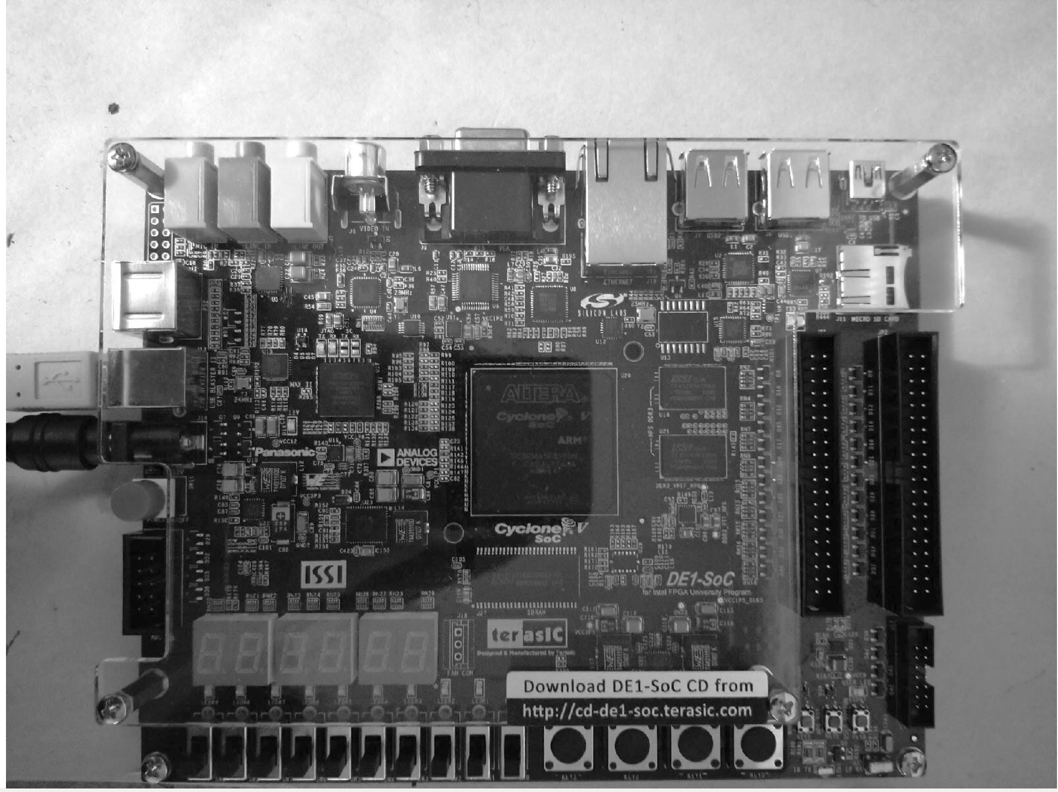
# Boundary of. Region

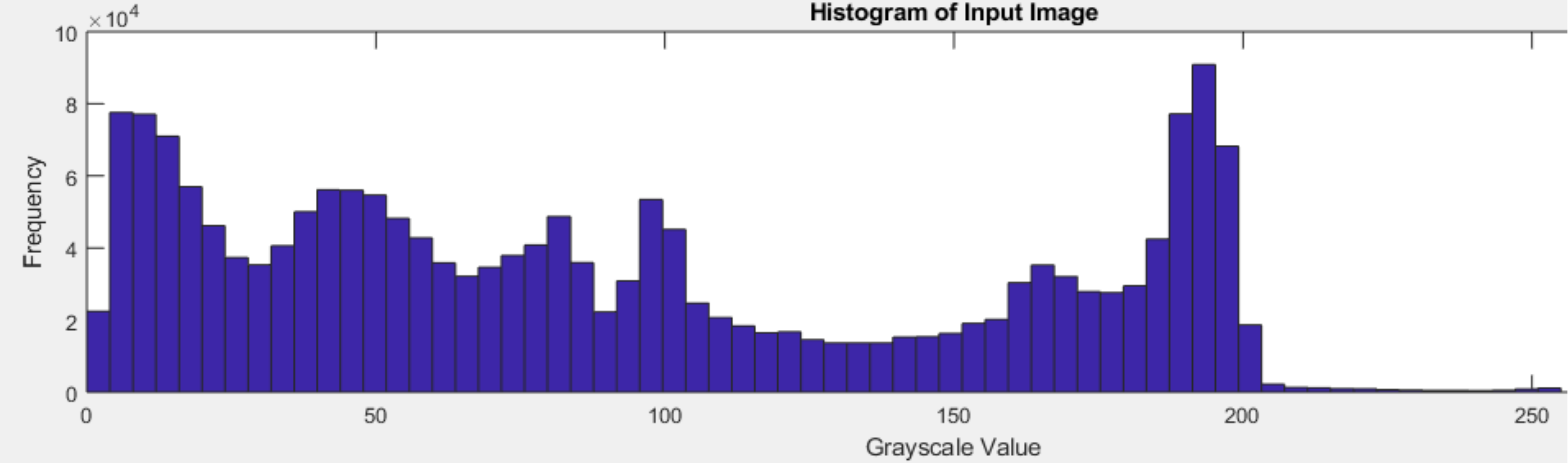
- Boundary of region R
  - Set of pixels within R such that each pixel has a neighbor that is NOT in R
  - Which pixels in green region are boundary pixels ?
    - Assuming 8 connectivity ?
    - Assuming 4 connectivity ?
  - Which pixels in red region are boundary pixels ?



# Image Histogram

- **Image histogram** = histogram of intensity values in a grayscale image
  1) Bin/bucket range of values:
     divide range into a series of intervals
  2) Count number of values in each interval
     - Sometimes count scaled to model probability

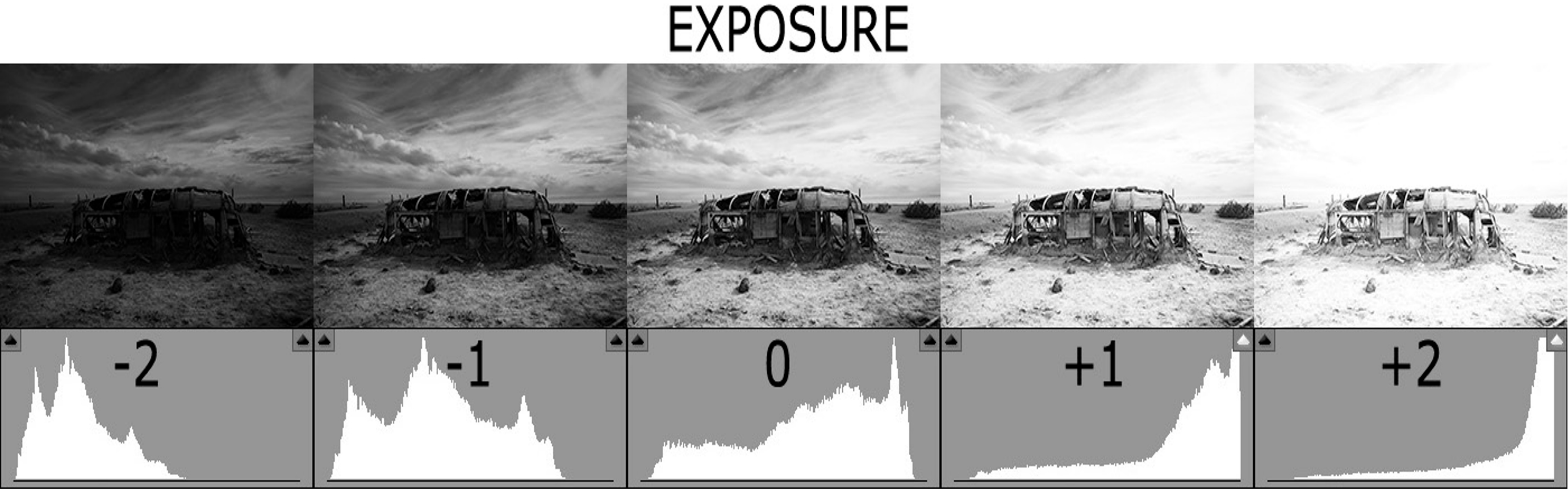


# Image Histogram

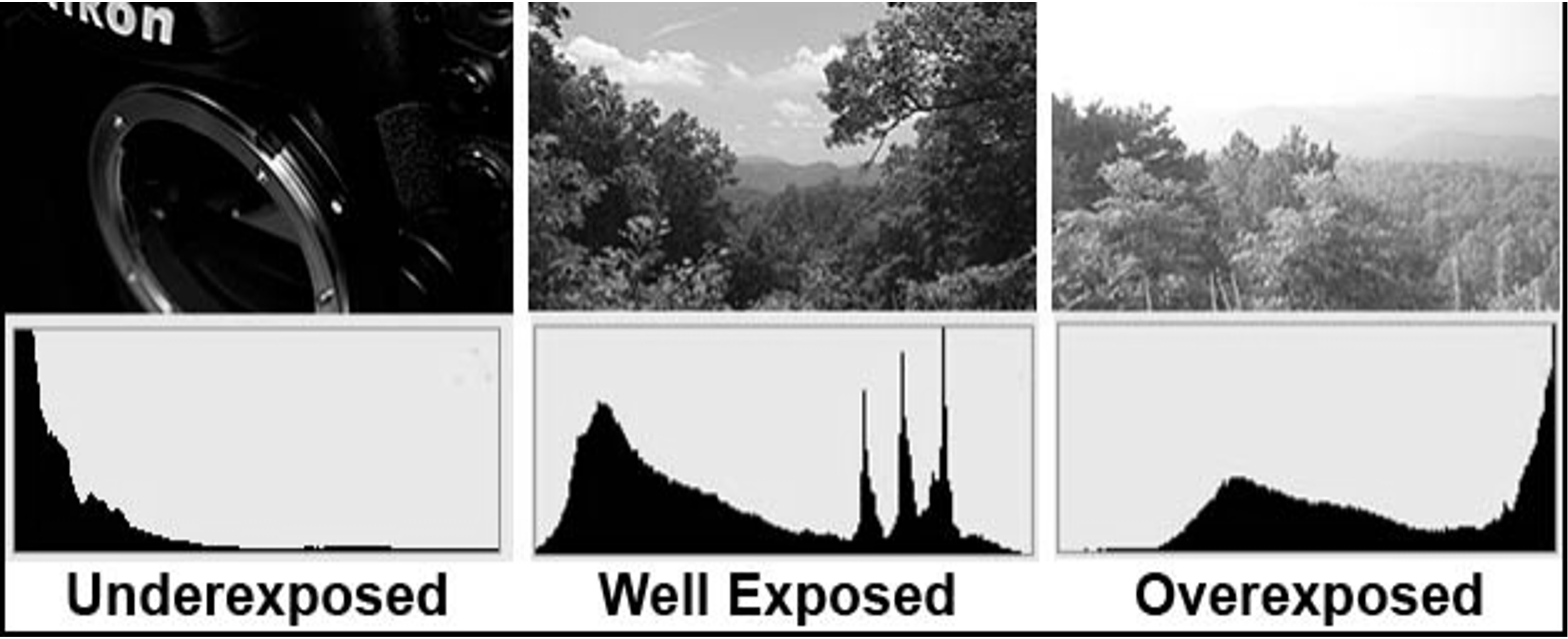
- Histograms are related to photograph exposure

1) Aperture size (amount of light per unit time)
2) Shutter speed (time of exposure)



# Image Histogram

- Histograms are related to photograph exposure



# Point Operations on Grayscale Images

- We'd like transformations T(.) to be monotonic functions

- Monotonically increasing function: $a > b \rightarrow T(a) \geq T(b)$
- Under monotonically-increasing transformations:
  if,
  before transformation, pixel at x is brighter than pixel at y
  then,
  after transformation, pixel at x cannot become darker than pixel at y

- Monotonically decreasing function: $a < b \rightarrow T(a) \leq T(b)$

# Point Operation: Image Thresholding

- **Global thresholding** (for real-valued image)
  - If intensity g < $\tau$: then T(g) = u (some constant)
    If intensity g ≥ $\tau$: then T(g) = v (some constant)


# Point Operation: Image Thresholding

- Global thresholding (for vector-valued image, e.g., RGB image)
  - If norm(c) < $\tau$: then T(c) = u (some constant)
    If norm(c) ≥ $\tau$: then T(c) = v (some constant)

# Otsu Thresholding

IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS, VOL. SMC-9, NO. 1, JANUARY 1979

## A Threshold Selection Method from Gray-Level Histograms

NOBUYUKI OTSU

*Abstract*—A nonparametric and unsupervised method of automatic threshold selection for picture segmentation is presented. An optimal threshold is selected by the discriminant criterion, namely, so as to maximize the separability of the resultant classes in gray levels. The procedure is very simple, utilizing only the zeroth- and the first-order cumulative moments of the gray-level histogram. It is straightforward to extend the method to multithreshold problems. Several experimental results are also presented to support the validity of the method.

# Otsu Thresholding

- Let histogram have L bins with values { p(i) :i=0,…,L-1 } where $\sum_i$ p(i) = 1
- Any threshold (bin index) "t" splits histogram/image into 2 regions (labeled 0 and 1)

$$\omega_0(t) = \sum_{i=0}^{t-1} p(i)$$

- Let fraction of pixels in each region be $w_0$ and $w_1$, where $w_0 + w_1 = 1$

$$\omega_1(t) = \sum_{i=t}^{L-1} p(i)$$

- Otsu finds threshold "t" to **minimize** "within-class variance" = weighted sum of variances of intensities within 2 resulting regions

$$\sigma_w^2(t) = \omega_0(t)\sigma_0^2(t) + \omega_1(t)\sigma_1^2(t)$$

  - Weights = $w_0$ and $w_1$
- Exhaustive search over bin-end values